



VisaPath

AI Immigration Timeline Planner for International Students

Never miss a visa deadline again. VisaPath takes your specific situation and builds a personalized roadmap from OPT all the way to green card — powered by AI, grounded in real USCIS rules.

1M+

STUDENTS IN THE US

\$300/hr

LAWYER COST

56

USCIS RULES ENCODED

15+

ACTIVE USERS AT RIT

DEV DASH 2026

Built by Dileep Kumar Sharma | Rochester Institute of Technology

visapath-app.azurewebsites.net • github.com/Dileep2896/visapath

THE PROBLEM

Immigration info is scattered, confusing, and high stakes

There are over **1 million international students** in the US, and every one of them navigates a maze of visa deadlines, work authorization rules, and immigration paperwork. Miss one deadline and you could lose your legal status.

OPT Filing Window

90 days before graduation to 60 days after. Miss it by a single day and you lose work authorization. No second chance.

CPT Overuse Trap

Using 12+ months of full-time CPT permanently kills your OPT eligibility. Most students don't learn this until it's too late.

Country Backlogs

India and China nationals face 10–30+ year green card waits in EB-2/EB-3. This dramatically affects long-term planning.

Scattered Information

Students piece together info from USCIS.gov, Reddit, university DSO offices, and \$300+/hr immigration lawyers.

THE SOLUTION

One personalized timeline for your entire immigration journey

You fill out a short onboarding form — visa type, degree, STEM status, graduation date, country, career goal — and VisaPath generates a complete immigration roadmap tailored to your exact situation.

Smart Timeline

AI generates every deadline and milestone, color coded by urgency, grouped by month, with expandable action items.

Risk Alerts

Flags CPT overuse, country backlogs, H-1B lottery odds, wage-level weighting, and approaching deadlines.

AI Chat (RAG)

Ask any immigration question. Answers grounded in 8 USCIS knowledge base documents — not hallucinated.

Document Tracker

Step-by-step checklists for OPT, STEM OPT, H-1B, and Green Card filings with document tracking.

Tax Guide

Personalized tax filing guidance based on visa type, country, treaty benefits, and residency status.

What-If Simulator

Change graduation date, STEM status, or country and instantly see how your entire timeline shifts.

HOW IT WORKS

Architecture & Timeline Generation

Timeline Generation Pipeline

- 1 **Student fills onboarding form** — visa type, degree, STEM status, graduation date, country, career goal
- 2 Backend assembles prompt with **56 hardcoded USCIS rules**, filing fees, H-1B lottery stats, country backlogs
- 3 **Gemini 2.5 Flash** generates structured JSON timeline grounded in real rules, not training data
- 4 **Risk engine** analyzes for CPT overuse, country backlogs, wage-level lottery odds
- 5 Frontend renders **interactive timeline** with urgency colors, expandable cards, month grouping

RAG Chat Pipeline

- 1 **8 immigration docs** split into 21 chunks, embedded with gemini-embedding-001 (768 dim)
- 2 Query embedded and matched via **cosine similarity** in ChromaDB (top k=4)
- 3 Retrieved chunks + user visa context sent to **Gemini for a grounded answer**

Key Design Decision

We don't rely on AI training data for immigration rules. 56 hardcoded USCIS rules are injected into every prompt as ground truth.

System Architecture

```
React 19 + TypeScript + Tailwind CSS v4 (Frontend SPA)
  | REST API (JSON over HTTPS)
  v
Python 3.13 + FastAPI (Backend)
  |— Timeline Generator —> 56 USCIS Rules + Fees + Backlogs —> Gemini 2.5 Flash
  |— Chat Service —> ChromaDB RAG (21 chunks) —> Gemini 2.5 Flash
  |— Tax Guide Service —> RAG Context —> Gemini 2.5 Flash
  |— Auth Service —> JWT + bcrypt —> PostgreSQL
  |— Risk Analyzer —> Immigration Rules + Country Data
```

Risk Analysis Engine

RISK	SEVERITY	TRIGGER
CPT overuse (12+ months)	Critical	Permanently kills OPT eligibility
OPT deadline approaching	Critical	<30 days to filing window close
Country backlog (India/China)	Warning	10–30+ year green card waits
H-1B wage-level weighting	Warning	Lower odds for entry-level under FY2027+ rules
Non-STEM limited OPT	Info	Only 12 months, no STEM extension
H-1B lottery uncertainty	Info	~25–30% selection rate

TECH STACK

Built With

LAYER	TECHNOLOGY
Frontend	React 19, TypeScript, Vite 7
Styling	Tailwind CSS v4 (custom dark theme)
Routing	React Router v7
Backend	Python 3.13, FastAPI
AI Model	Google Gemini 2.5 Flash
Embeddings	gemini-embedding-001 (768 dim)
Vector DB	ChromaDB + LangChain
Auth	JWT + bcrypt
Database	SQLite (dev) / PostgreSQL (prod)
Hosting	Azure App Service (Student tier)
CI/CD	GitHub Actions

Key Technical Decisions

- **56 hardcoded USCIS rules** injected into every AI prompt — we don't rely on model training data
- **RAG pipeline** with 8 docs, 21 chunks, cosine similarity — answers cite real USCIS sources
- **Client-side date recalculation** — cached timelines recalculate urgency on every render
- **Fire-and-forget navigation** — animated loading turns 15s AI wait into engaging experience
- **Lazy-loaded routes** — main bundle reduced from 483KB to 270KB
- **4-layer rate limiting** — per-IP, daily AI tracker, frontend pre-check, Gemini 429 cooldown

SCREENSHOTS

App Interface

VisaPath

Timeline

Alerts

Action Items

AI Chat

Documents

Tax Guide

Profile

What-If Scenarios

Current Status

F-1

Student (OPT/On-Campus)

2

Recommended OPT Application Date

As a citizen of India, you face severe backlogs for employment-based Green Cards (EB-2 and EB-3 categories), with estimated wait times ranging from 10 to 30 years. This means even after your I-140 is approved, you could wait decades for your Green Card.

Your employer is not cap-exempt, meaning you must go through the H-1B lottery. The H-1B lottery is highly competitive, and selection is not guaranteed, even with a Master's degree.

For the FY2027 H-1B lottery and beyond, USCIS uses a wage-level weighted selection system. Your expected wage level (Level 1-Entry) will result in a significantly lower selection odds (approximately 45% lower) compared to higher wage levels.

View all 4 alerts

Your Timeline

Show 20 past events

FEBRUARY 2026

Recommended OPT Application Date

Due Date: Feb 15, 2026

Review

MARCH 2026

H-1B FY2027 Lottery Registration Opens

Due Date: Mar 1, 2026

In 6 days

H-1B FY2027 Lottery Registration Deadline

Due Date: Mar 15, 2026

In 1 month

APRIL 2026

H-1B FY2027 Lottery Results Announced

Due Date: Apr 1, 2026

In 1 week

MAY 2026

Expected Graduation Date

Due Date: May 5, 2026

In 3 months

Estimated OPT Start Date

Due Date: May 15, 2026

In 4 months

Timeline Dashboard

VisaPath

Timeline

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Risk Alerts

1 Critical

2 High Priority

1 Warning

CRITICAL

RISK ALERTS

As a citizen of India, you face severe backlogs for employment-based Green Cards (EB-2 and EB-3 categories), with estimated wait times ranging from 10 to 30 years. This means even after your I-140 is approved, you could wait decades for your Green Card.

Examine options for EB-1 (Extraordinary Ability, Outstanding Researcher/Professor, Multinational Manager) or EB-2 (National Interest Waiver (NIW)) categories, which typically have shorter wait times for Indian nationals. Consult with an immigration attorney to assess your eligibility for these categories early in your career.

HIGH PRIORITY

RISK ALERTS

Your employer is not cap-exempt, meaning you must go through the H-1B lottery. The H-1B lottery is highly competitive, and selection is not guaranteed, even with a Master's degree.

Prepare for the possibility of not being selected in the H-1B lottery. Discuss alternative visa options or long-term strategies with your employer, such as O-1, L-1, or exploring cap-exempt employment if H-1B is critical.

For the FY2027 H-1B lottery and beyond, USCIS uses a wage-level weighted selection system. Your expected wage level (Level 1-Entry) will result in a significantly lower selection odds (approximately 45% lower) compared to higher wage levels.

Discuss with your employer the possibility of targeting positions that qualify for higher wage levels (Level 4, II, or IV) in the H-1B lottery process, as this could potentially increase your chances of selection. Also, consider roles that might qualify for O-1 or EB-1A if applicable to your profile.

WARNING

RISK ALERTS

You have used 10 months of full-time OPT. Using 10 or more months of full-time OPT makes you ineligible for Optional Practical Training (OPT). You are currently close to this limit.

Ensure you do not exceed 12 months of full-time OPT. Carefully track your OPT usage and consult your DSO if you have any doubts about your eligibility for OPT.

Risk Alerts

VisaPath

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AI Immigration Advisor

Answers grounded in official USCIS documentation

Research official USCIS website or use our secure private chat interface to ask questions based on official USCIS documentation.

What happens if I exceed 90 days of unemployment on OPT?

ⓘ

Navigating OPT unemployment rules is critical for maintaining your F-1 status, and it's wise to understand these limits as a Master's student on a STEM field from India. For your initial 12-month Optional Practical Training (OPT), the rule states:

- Unemployment Limit:** You are permitted a maximum of **90 days of unemployment** during your 12-month OPT period.
- Every period of unemployment contributes to this 90-day limit.
- Consequence of Exceeding this Limit:** If you exceed these 90 days of unemployment, your **F-1 status will be automatically terminated**. This is a serious consequence, as it means you lose your legal status to remain in the US, and typically means you must depart the country immediately, as there is no grace period if your status is terminated.

Linking Ahead to STEM OPT (Relevant for you as a STEM Master's student):
If you later apply for and are approved for the STEM OPT Extension, the unemployment rules change slightly to cover the entire 36-month period (Initial 12-month OPT + 24-month STEM OPT extension):

- Aggregate Unemployment Limit:** For the entire 36-month period of OPT and STEM OPT combined, you are allowed a maximum of **180 days of aggregate unemployment**. This limit includes any unemployment days accumulated during your initial 12-month OPT.
- Consequence of Exceeding the Aggregate Limit:** Similar to the initial OPT, exceeding the 180 days of aggregate unemployment across your entire 36-month period will also result in the termination of your F-1 status.

Important Considerations and Recommendations:

- Reporting is Key:** You must report your employment information (employer name, address, any changes, or interruption) to your Designated School Official (DSO) within 10 days of the event. This allows your DSO to accurately track your unemployment days.
- Proactive Planning:** It's crucial to actively search for employment and be mindful of your unemployment clock. Many students plan their OPT start date carefully to align with potential job offers.
- No Work Without EAD:** Remember, you cannot begin working until you have received your EAD card and on or after the start date printed on it.
- Consult Your DSO:** Given the serious implications of exceeding unemployment limits, always consult with your DSO at your academic institution if you are approaching the limit or have any concerns about your employment situation. They are your primary resource for advice tailored to your specific situation and can help you understand all reporting requirements and potential options.

This information is based on USCIS regulations for F-1 students on OPT. Always ensure you are following the most current guidelines and report any changes promptly to your DSO.

ⓘ Based on official USCIS documentation

Ask about your immigration situation.

AI Chat (RAG)

VisaPath

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Document Tracker

Track what you need for each immigration step

OPT Application

STEM OPT Extension

H-1B Petition

Green Card (PERM)

2 of 8 documents ready

25%

✓ Forms I-20B

Application for Employment Authorization

USCIS website (the online e-file tool)

○ Updated - 20

With OPT extension from your DSO on page 2

of Your university (DSO) office

○ Passport Photos

2x 2x2 inch photos (2x2 inches, white background)

of CIVS, Waiver, or online passport photo service

✓ I-14 Record

Most recent arrival/departure record

of I-94.dsp.dsp.dsp

○ Passport Copy

Photo page and US visa stamp page

of your passport

○ Previous EAD Cards

Copies of all previous Employment Authorization Documents

of your records

○ All Previous I-20s

Copies of all I-20 forms from current and previous programs

of your records

○ Filing Fee

1,765 filing fee (\$40 as of 2024, check current amount)

of USCIS fee schedule page

Document Tracker

VisaPath

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Tax Filing Guide

Personalized for your immigration profile

This guide is generated once per profile to conserve AI usage. Update your profile to get a fresh guide.

FILING STATUS

Filing Deadline: April 15, 2026

Residency Status: Nonresident Alien

REQUIRED FORMS

Form 9843

Form 1040-NR

FICA TAX EXEMPTION

You are currently EXEMPT from Social Security and Medicare taxes (FICA). If your employer withholds these taxes, you can request a refund.

Exempt

PERSONALIZED GUIDANCE

Filing Requirements

As an F-1 visa holder from India in your second calendar year in the US, you are considered a **Nonresident Alien** for federal tax purposes. All F-1 visa holders present in the US during the tax year must file a federal tax return or information statement by **April 15**. Since you had **US-source wage income**, you are required to file an income tax return.

Key Forms

Based on your profile, you will likely need to file the following forms with the IRS:

- Form 9843, Statement for Exempt Individuals:** This form is mandatory for all Nonresident Aliens, including F-1 students, even if you had no income. It establishes your exempt individual status under the Substantial Presence Test.
- Form 1040-NR, U.S. Nonresident Alien Income Tax Return:** Since you earned **US-source wage income** as a Nonresident Alien, you must file this form to report your taxable income.
- You should receive a **Form W-2, Wage and Tax Statement**, from your employer by January 31, detailing your wages and taxes withheld. This form is used to prepare your 1040-NR.

Treaty Benefits

The United States has income tax treaties with over 70 countries that can reduce or eliminate tax on certain types of income for students. While India may have a tax treaty with the US, the provided reference documents do not specify the exact benefits available for students from India. If applicable, you would typically use **Form 8833** with your employer to claim exemption from withholding on personal services income based on a treaty. It is recommended to consult official IRS publications or a tax professional to determine if you are eligible for any treaty benefits.

FICA Status

As an F-1 student in your second calendar year in the US, you are **exempt from FICA taxes** (Social Security and Medicare taxes) for your first five calendar years. This exemption applies to your employer, OPT, and CPT employment. You should verify your **Form W-2** to ensure that FICA taxes were not incorrectly withheld from your wages. If they were, you may be able to request a refund using **Form 843**.

Tax Guide

VisaPath

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Action Items

Your prioritized to-do list based on upcoming deadlines

0 of 41 completed

0%

○ File Form I-765 (Application for Employment Authorization) with USCIS - filing fee: \$860

Due: Feb 15, 2026

Recommended OPT Application Date

Urgent

○ Include your new OPT I-20 from your DSO, passport photos, copy of passport, visa, I-94, and previous EADs (if any)

Due: Feb 15, 2026

Recommended OPT Application Date

Urgent

○ Keep copies of all submitted documents for your records.

Due: Feb 15, 2026

Recommended OPT Application Date

Urgent

○ Confirm with your employer that they will register you for the H-1B lottery.

Due: Mar 1, 2026

H-1B FY2027 Lottery Registration Opens

Soon

○ Provide your employer with all necessary personal information for registration.

Due: Mar 1, 2026

H-1B FY2027 Lottery Registration Opens

Soon

○ Ensure your employer pays the H-1B registration fee: \$235 per beneficiary.

Due: Mar 1, 2026

H-1B FY2027 Lottery Registration Opens

Soon

○ Follow up with your employer to confirm successful registration submission.

Due: Mar 31, 2026

H-1B FY2027 Lottery Registration Deadline

Upcoming

○ Keep a record of your H-1B registration confirmation number.

Due: Mar 31, 2026

H-1B FY2027 Lottery Registration Deadline

Upcoming

○ Await notification from your employer regarding your lottery selection status.

Due: Apr 1, 2026

H-1B FY2027 Lottery Results Announced

Upcoming

○ If selected, begin preparing documents for the H-1B petition filing with your employer.

Due: Apr 1, 2026

H-1B FY2027 Lottery Results Announced

Upcoming

○ Confirm your official graduation date with your university's registrar.

Due: May 5, 2026

Expected Graduation Date

Upcoming

○ Obtain your final transcript and diploma once available.

Due: May 5, 2026

Expected Graduation Date

Upcoming

○ Ensure your DSO updates your SEVIS record to reflect your graduation.

Due: May 5, 2026

Expected Graduation Date

Upcoming

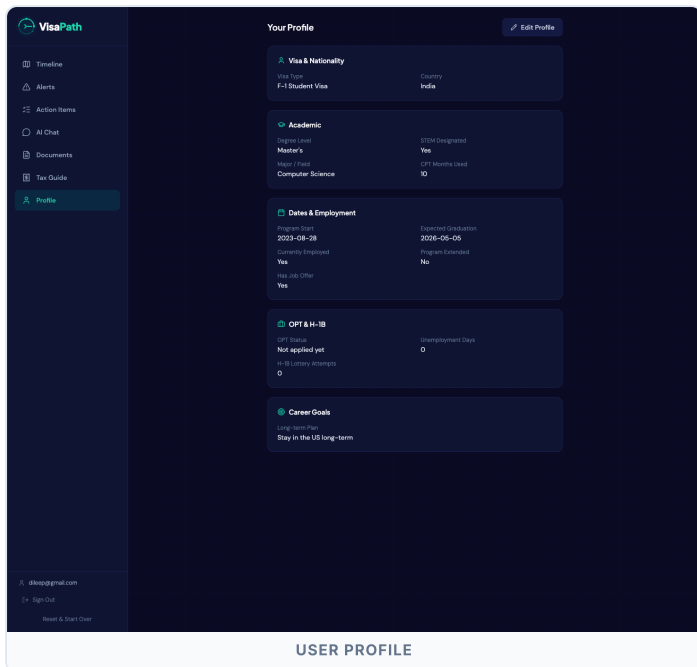
○ Do not begin employment until you have received your OPT EAD card.

Due: May 6, 2026

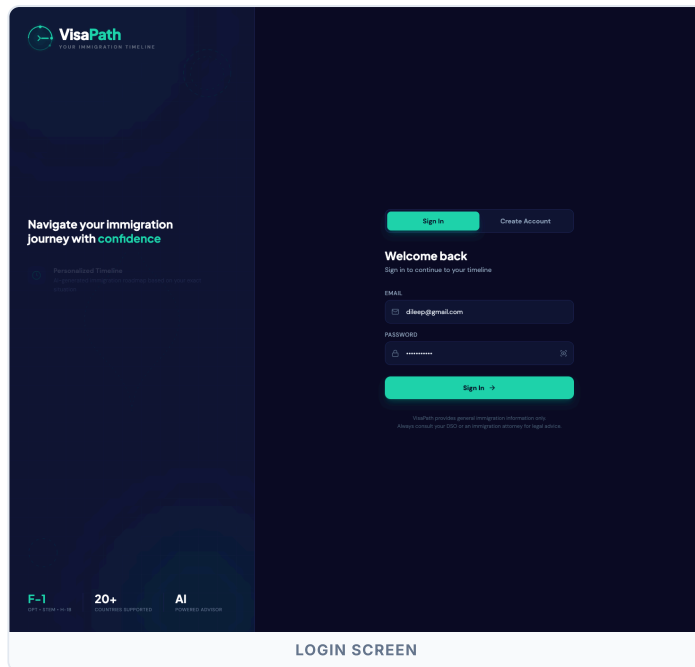
Estimated OPT Start Date

Upcoming

Action Items



USER PROFILE



LOGIN SCREEN

TRACTION

Real Students, Real Impact

We shared VisaPath with **15 classmates at RIT**, all F-1 international students. Their feedback validated that the tool catches things even knowledgeable students miss.

"I knew I had to file OPT before graduation, but VisaPath calculated my exact filing window down to the day and showed me I only had 23 days left. I was planning to file next month — that would have been too late."

Priya Sharma — MS Data Science, F-1

"I used the what-if simulator to compare staying on OPT vs applying for H-1B through a cap-exempt employer. It showed me a path I didn't even know existed — skipping the lottery entirely through a university-affiliated research position."

Ananya Reddy — MS Bioinformatics, F-1

"The tax guide told me India has a US tax treaty for students that most people don't claim. It also showed the exact forms I need as a dual-status filer."

Siddharth Joshi — MS Computer Science, F-1

"The risk engine flagged that the H-1B lottery is switching to wage-level weighting in FY2027. VisaPath laid out alternate paths like the O-1B and EB-2 NIW that I hadn't considered."

Vikram Singh — MBA, F-1

How VisaPath Becomes Sustainable

TIER	PRICE	INCLUDES
Free	\$0	5 timeline generations, basic risk alerts, document checklists
Pro	\$9.99/mo	Unlimited timelines, AI chat, what-if simulator, tax guide, push notifications
University	\$499/yr	DSO advisor dashboard, bulk student management, analytics, priority support

TAM: 1M+ students

~400K new F-1 students enter the US each year. Average spend on immigration help: \$500-2K.

0 direct competitors

No existing tool provides AI-powered personalized immigration timelines for students.

Prize Money Investment Plan — \$5,000

AMOUNT	USE	DETAILS
\$2,000	User acquisition at 10 universities	Campus ambassadors, student org partnerships, DSO outreach
\$1,500	AWS production infrastructure	Always-on ECS, RDS PostgreSQL, CloudFront CDN — 12 months
\$1,000	AI API costs	Gemini API for timeline, chat, tax guide at scale for 1,000+ users
\$500	Legal review	Immigration attorney validates our 56 USCIS rules and disclaimers

\$5/user customer acquisition cost — 12 months of runway to reach 1,000 active users

Future Roadmap

- Push notifications for upcoming deadlines (email + browser)
- Support for J-1, L-1, O-1 visa types
- Cap-exempt employer lookup database
- Live USCIS processing time integration
- University DSO advisor dashboard
- React Native mobile app (iOS + Android)
- AWS migration for production-grade hosting
- Partnerships with university international offices

Runtime AI (in the product)

Google Gemini 2.5 Flash powers timeline generation, AI chat Q&A, and tax guide. Google gemini-embedding-001 creates vector embeddings for the RAG pipeline.

Development AI (building the product)

Claude Code (Anthropic) was used as a coding assistant during the hackathon for code generation, debugging, and iteration.

All immigration guidance in the app includes disclaimers that it's general information, not legal advice. Users are always directed to consult their DSO or an immigration attorney.

